

Audio Representation

Learning Objectives

- Understand how sound waves become digital data
- Explain sampling rate and bit depth in audio
- Calculate file size for different audio quality settings

Engage (5-7 minutes)

Think of taking measurements of river water level every hour versus every minute—more frequent measurements give a clearer picture of changes. Audio sampling works the same way—measuring sound wave height many times per second.

Explore (10-12 minutes)

Use Audacity to record a 5-second clip. Check the project rate—usually 44100 Hz. Record another at 8000 Hz. Compare file sizes and sound quality. Write which sounds clearer and which file is smaller.

Explain (10-12 minutes)

Analog sound waves are continuous. Digital audio captures snapshots called samples—thousands per second. Sampling rate is samples per second—44100 Hz means 44,100 measurements each second. Bit depth is sample precision—16-bit means 65,536 possible levels. Higher rate and depth mean better quality but larger files.

Elaborate (8-10 minutes)

Calculate file size for 1 minute of audio: $44100 \text{ samples/sec} \times 16 \text{ bits} \times 60 \text{ sec} = 42,336,000 \text{ bits}$. Convert to bytes, then megabytes. Repeat for 8000 Hz and 8-bit audio. Create a table comparing quality and file sizes.

Evaluate (5-8 minutes)

Exit ticket: Define sampling rate and bit depth. Explain why CD-quality audio uses 44100 Hz sampling rate.